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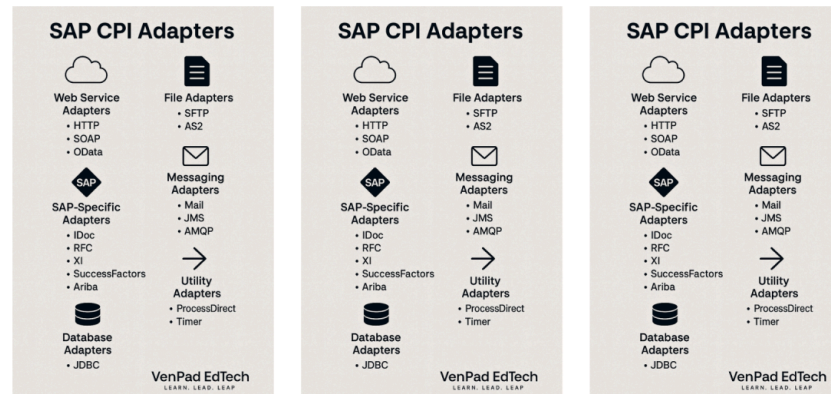
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Complete Guide to SAP CPI Adapters

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SAP Cloud Platform Integration (SAP CPI) provides a rich set of adapters that enable communication between various systems, applications, and protocols. These adapters act as the entry and exit points in an iFlow and determine how messages are sent or received. This guide provides an in-depth look at all the major SAP CPI adapters, categorized by usage type.

Web Service and API Adapters

1. HTTP Adapter

- **Direction:** Sender & Receiver
- **Use:** REST API integration, webhooks, external HTTP services
- **Features:** Supports GET, POST, PUT, DELETE, OAuth2, Basic Auth, CSRF token handling
- **Example:** Integrating with third-party APIs like Salesforce, payment gateways

2. SOAP Adapter

- **Direction:** Sender & Receiver
- **Use:** Legacy system integration using SOAP
- **Features:** WSDL import, SOAP 1.1/1.2, Basic Auth, Certificate Auth
- **Example:** Communicating with banking systems or legacy CRM

3. OData Adapter

- **Direction:** Receiver
- **Use:** Accessing SAP S/4HANA, SuccessFactors OData services
- **Features:** Supports OAuth2, CRUD operations, \$filter/\$expand
- **Example:** Fetching employee records from SuccessFactors

4. REST Adapter (via HTTP)

- **Note:** No dedicated REST adapter, implemented using HTTP Adapter
- **Use:** Lightweight REST APIs with JSON or XML payloads

5. Open Connectors Adapter

- **Direction:** Receiver
 - **Use:** Unified connection to third-party SaaS apps
 - **Example:** Google Sheets, Zendesk, HubSpot
-

File-Based Adapters

6. SFTP Adapter

- **Direction:** Sender & Receiver
- **Use:** Secure file exchange with external systems
- **Features:** SSH Key, polling, file processing modes, timezone settings
- **Example:** Sending invoices to a vendor's SFTP server

7. FTP Adapter (Deprecated)

- **Note:** Use SFTP instead due to enhanced security

8. AS2 Adapter

- **Direction:** Sender & Receiver
- **Use:** Secure B2B file transfers (e.g., EDI)
- **Features:** Encryption, signing, MDN support, certificate management
- **Example:** Exchanging purchase orders with a trading partner

9. EDI Separator Adapter

- **Use:** Splits bulk EDI messages (X12, EDIFACT) into individual messages
 - **Features:** Control Number handling, envelope splitting
-

Messaging and Communication Adapters

10. Mail Adapter

- **Direction:** Sender & Receiver
- **Use:** Trigger flows via email or send notifications
- **Features:** SMTP, IMAP, POP3, attachments, subject filtering
- **Example:** Receiving HR requests from mail, sending approval notifications

11. AMQP Adapter

- **Use:** Publish/subscribe messaging (via RabbitMQ, etc.)
- **Example:** Event-driven architecture

12. JMS Adapter

- **Use:** Asynchronous message queuing via SAP Event Mesh
 - **Example:** Decoupled integration of microservices
-

SAP-Specific Adapters

13. IDoc Adapter

- **Direction:** Sender & Receiver
- **Use:** Communicate with SAP ECC/S4 via IDocs
- **Features:** Logical system, partner profile config, SAP Cloud Connector required
- **Example:** Create Sales Order from external source

14. RFC Adapter

- **Use:** Remote Function Call with SAP on-prem systems
- **Example:** Call BAPI to fetch customer master data

15. XI Adapter

- **Use:** Communicate with SAP PI/PO using XI 3.0 protocol
- **Example:** Forward messages to existing PI system

16. SuccessFactors Adapter

- **Use:** Access SF OData v2 or SFAPI
- **Features:** OAuth2, entity filter, query options
- **Example:** Extract employee job data

17. Ariba Adapter

- **Use:** Exchange procurement data with SAP Ariba

18. SAP Cloud for Customer (C4C) Adapter

- **Use:** Connect to SAP C4C
- **Example:** Sync leads or service tickets

19. SAP Business ByDesign Adapter

- **Use:** Integration with SAP ByD suite

20. SAP S/4HANA Cloud Adapter

- **Use:** Special handling for S/4HANA Cloud APIs

Database and Data Service Adapters

21. JDBC Adapter

- **Use:** Read/write to on-prem DBs (MySQL, Oracle)
- **Requirement:** SAP Cloud Connector, JDBC driver, on-prem agent
- **Example:** Fetch orders from MySQL for processing

22. OData for HANA Adapter (Part of OData)

- **Use:** Consume HANA-based OData APIs

Internal / Utility Adapters

23. Process Direct Adapter

- **Use:** Call one iFlow from another
- **Features:** Fast, no network hops, modular design

24. Timer Adapter

- **Use:** Scheduled flow triggers
- **Example:** Daily report generation at 9 AM

25. HTTPS Adapter

- **Use:** Secure inbound calls with custom certificates

26. Generic Adapter

- **Use:** Extensibility or custom built integration logic

Authentication Mechanisms

- **Basic Authentication:** HTTP, SOAP, JDBC, OData
- **OAuth2:** HTTP, OData, SuccessFactors
- **Certificate-Based:** SOAP, AS2, SFTP
- **SAML:** OData for S/4HANA Cloud

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